

Tagging and Parsing

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Introduction	325
Definitions of Tagging and Parsing	326
Tagging	326
Parsing	326
Automated Tagging and Parsing	326
Effects of Particular Linguistic Features	327
Effects of Linguistic Domain of Training Data	328
Effects of the Amount of Training Data	328
Effects of the Machine Learning Model Used	328
Other Types of Automatic Linguistic Annotation	328
Challenges in Tagging and Parsing Low-Resource and Learner Language	328
Conclusion	329
References	329

Key Points

- Part-of-speech (POS) tagging assigns word-class labels, whereas syntactic parsing derives constituency or dependency structure; together they form the backbone of much corpus linguistic inquiry.
- Off-the-shelf NLP toolkits are overall highly accurate, though accuracy often drops for infrequent tags and out-of-domain language varieties.
- Performance hinges on the frequency and ambiguity of the target features, the match between training and target domains, the size of the annotated corpus, and the sophistication of the model architecture.
- Transformer-based pre-training now supports other kinds of automatic linguistic annotation (e.g., authorial stance, argument structure constructions) with relatively small training datasets, but learner varieties still require targeted annotation and domain-specific fine-tuning to meet research thresholds.

Abstract

In this chapter we introduce part of speech (POS) tagging and syntactic parsing. We define important terms and provide examples of tagging and parsing formalisms. We then discuss common uses for taggers and parsers in corpus linguistic research. Finally, we discuss important issues that should be considered when using automated linguistic annotation tools.

Introduction

Corpus linguistics is a field of language research that involves computer-aided analyses of large collections of machine-readable language data. Much corpus linguistic research has relied on texts that have been annotated for grammatical and syntactic features. Given the scale at which modern corpus linguistic analyses are often conducted (e.g., hundreds of millions or even billions of words), manual annotation is usually infeasible. Automated grammatical taggers and syntactic parsers are therefore often used to annotate both written and spoken corpus texts (e.g., Biber et al., 2004; Granger & Bestgen, 2014; Römer et al., 2015) prior to linguistic analysis. In this chapter, we provide an introduction to part of speech tagging and syntactic parsing and discuss current issues in the use of automated taggers and parsers.

Definitions of Tagging and Parsing

Tagging

The term “tagging” primarily refers to the annotation of grammatical part of speech (POS) tags. POS tags can be broadly categorized into two categories: Large-grained and fine-grained tags. Large-grained tags represent distinctions between broad word classes. The Universal POS tagset (UPOS; Nivre et al., 2020, pp. 4034–4043), for example, includes 17 POS tags that distinguish between word classes such as adjective (ADJ), adverb (ADV), noun (NOUN), and verb (VERB), but do not provide many distinctions within each word class (though proper nouns are distinguished from common nouns). Fine-grained tags provide additional language-specific distinctions within those same POS categories. The Penn Treebank Tagset (Marcus et al., 1993; Santorini, 1990), for example, includes 36 fine-grained POS tags. Unlike UPOS, which includes a single tag for all common nouns, the Penn tagset includes two tags for common nouns (singular/mass [NN] and plural [NNS]). Furthermore, in contrast to UPOS’s single tag for lexical verbs, the Penn tagset includes six tags that represent distinct verb inflections (base form [VB], past tense [VBD], gerund/present participle [VBG], past participle [VBN], non-third person singular present [VBP], third person singular present [VBZ]). See Fig. 1 for an example sentence tagged with large-grained (UPOS) and fine-grained (Penn Treebank Tagset) tags. Other tagsets provide even finer distinctions. For example, the Biber Tagger (Biber, 1988; Biber et al., 2004) includes over 100 fine-grained tags, and the C7 tagset used by the Constituent Likelihood Automatic Word-tagging System (CLAWS) POS tagger (Garside, 1996) includes over 140 fine-grained tags. POS information has been used for a variety of purposes in corpus linguistic research. Some of these purposes include investigating linguistic variation across registers (e.g., Biber, 2006), analysis of multiword units (e.g., Granger & Bestgen, 2014), and homograph disambiguation (e.g., Jarvis & Hashimoto, 2021) among many others.

Parsing

The term “parsing” refers to the annotation of syntactic structure in a text. There have been two main approaches to syntactic annotation, namely constituency parsing and dependency parsing.

A syntactic constituency parse represents a sentence/utterance as a syntactic constituency tree (See Fig. 2 for an example sentence). While many formalisms are possible, the most common formalism in English treebanks has been the Penn Treebank syntactic tagset (e.g., Penn Treebank, Marcus et al., 1993; OntoNotes, Weischedel et al., 2013). The Penn Treebank syntactic tagset includes labels for 14 phrase and clause types such as simple declarative clauses (S), noun phrases (NP), and verb phrases (VP). Constituency parsed texts have been used for a number of research purposes. Lu (2010), for example, used syntactic constituency parsed texts to identify features of syntactic complexity (e.g., T-units, dependent clauses, complex nominal clauses) in a large corpus of learner texts.

A syntactic dependency parse represents a sentence/utterance as a series of functional grammatical relationships between syntactic dependents and their syntactic heads (see Fig. 3). While some formalisms are language specific (e.g., CLEARNLP for English; Choi et al., 2015), others are designed to be broadly applicable across languages (e.g., Universal Dependencies, Nivre et al., 2020). Generally speaking, dependency representations are more fine-grained than constituency representations, as they include the functional role between phrase and clause types. For example, the Universal Dependencies (UD) tagset includes seven distinct labels for noun phrases that are dependents of a verb phrase (e.g., nominal subject [nsubj], direct object [obj], oblique [obl]) and an additional five distinct labels for noun phrases that are dependents of other nouns (e.g., noun modifier [nmod], appositive [appos]). Dependency parsed texts have been used for a wide variety of purposes in corpus linguistic research. For example, researchers have relied on syntactic dependency labels to identify fine-grained features of syntactic complexity (e.g., Kyle & Crossley, 2018) or particular verb argument constructions (e.g., Kyle & Crossley, 2017; Römer et al., 2014).

Automated Tagging and Parsing

Although some early automated taggers and parsers relied on manually defined rules (e.g., Biber, 1988), most modern tools are automatically trained using a manually annotated dataset and a machine learning algorithm. While machine learning techniques

	1	2	3	5	6	8	9	10	11	12	13	14	15
<i>Token</i>	His	artworks	were	selected	and	exhibited	in	the	British	Museum	in	2002	.
<i>UPOS tags</i>	PRON	NOUN	AUX	VERB	CCONJ	VERB	ADP	DET	PROPN	PROPN	ADP	NUM	PUNCT
<i>Penn tags</i>	PRP\$	NNS	VBD	VBN	CC	VBN	IN	DT	NNP	NNP	IN	CD	.

Fig. 1 A comparison of UPOS tags versus Penn Treebank tags for a sentence sourced from the English Web Treebank (Bies et al., 2012; Silveira et al., 2014, pp. 2897–2904).

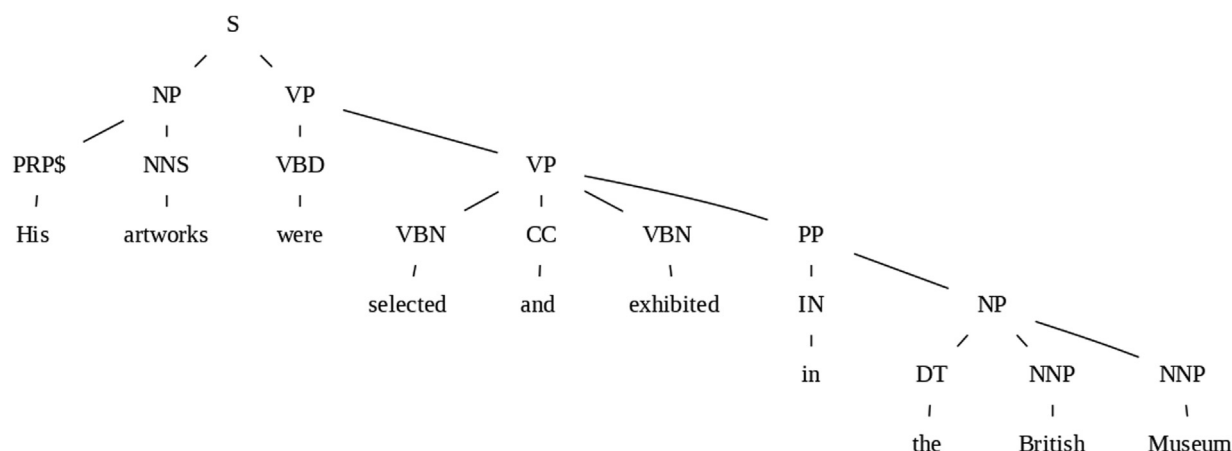


Fig. 2 An example of constituency parse tree, visualized using *spaCy* (Honribal et al., 2020) with the *benepar_en3* parser (Kitaev et al., 2019), wherein a sentence (S) is parsed into a noun phrase (NP) and a verb phrase (VP), which are further divided into their constituent parts. The initial NP, for example, consists of the possessive pronoun (PRP\$) *his* and a plural noun (NNS) *artworks*.

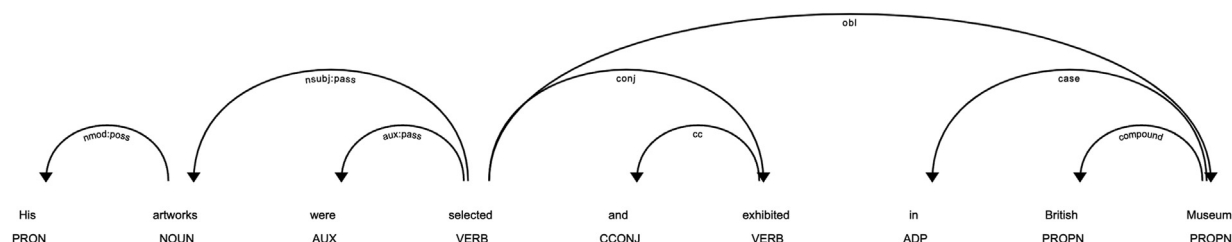


Fig. 3 An example of dependency parse tree with Universal Dependencies, visualized by *displaCy* (Honribal et al., 2020), wherein connections between words are expressed in the form of functional head/dependent relationships. For example, *artworks* functions as the passive subject (nsubj:pass) of the verb *selected*, and *Museum* functions as the head of a phrase that functions as an oblique (obl) of the verb *selected*.

constantly evolve, at the time of writing automated POS taggers achieve average accuracies as high as $F1^1 = 0.980$ (Nguyen & Verpoor, 2018, pp. 81–91) and parsers achieve average accuracies as high as $F1 = 0.960$ (Mrini et al., 2020).

There are many automated POS taggers and syntactic parsers available for use. Two increasingly popular options that include models for a range of target languages are Stanza (Qi et al., 2020, pp. 101–108) and *spaCy* (Honribal et al., 2020). Both tools provide automatic annotation of POS tags and syntactic dependency relations (among other annotation types) and are capable of achieving high accuracy rates (though this varies by particular model and language). Furthermore, both tools allow users to train customized annotation models. For a direct comparison of the performance of these two (and other) tools on L1 and L2 English written and spoken texts, see Kyle and Eguchi (2024). Although averaged accuracy figure for taggers and parsers are quite high, real-world results can vary for a number of reasons.

Effects of Particular Linguistic Features

While averaged accuracy figures provide a helpful point of comparison between models, these do not necessarily reflect the accuracy of a model for a particular grammatical annotation. Linguistic items that are rare in the training data and/or lexically/contextually ambiguous tend to be annotated with lower accuracy than those that are frequent in the training data and are less ambiguous. For example, Kyle and Eguchi (2024) trained a POS tagging model that achieved high average weighted accuracy figures ($F1 = 0.979$). Some tags were annotated with almost perfect accuracy (e.g., personal pronouns; $F1 = 0.997$), while others were annotated with much lower accuracy (e.g., comparative adverbs; $F1 = 0.867$). Practically speaking, this means that researchers need to know a tool's performance for the annotation type(s) of interest for a particular study and perform manual corrections when accuracy is below a pre-set threshold. Biber et al. (2025), for example, performed manual corrections until accuracy (precision) reached 95%.

¹F1 scores are a general indication of annotation accuracy. F1 scores represent the harmonized mean of *precision* ((number of true positives)/(number of true positives + false positives)) and *recall* ((number of true positives)/(number of true positives + false negatives)) and is calculated as $2 * ((precision + recall)/(precision + recall))$. F1 scores range from 0 to 1.0.

Effects of Linguistic Domain of Training Data

Automated taggers and parsers rely on the regularity of linguistic patterns in a particular dataset. When studies report the accuracy figures for a tagging and/or parsing model, they are usually evaluated on data that closely matches the training data (in terms of language variety, genre, register, and annotation conventions) rather than on truly out-of-domain text. For example, if a tagger/parser is trained on the publicly available UD version of English Web Treebank (Bies et al., 2012; Silveira et al., 2014, pp. 2897–2904) accuracy figures will be reported for a section of that treebank that is held out for the purposes of model testing (the test set). While the model is not trained using the test set, the linguistic domain is exactly the same. When this model is applied to datasets with different linguistic domains, lower performance may be exhibited (e.g., Kyle et al., 2022a, pp. 39–45; Plank & Van Noord, 2011, pp. 1566–1576). For example, Kyle et al. (2022a, pp. 39–45) reported the performance of a POS tagging model that was trained on four English treebanks consisting primarily of texts from the internet. The results demonstrated that the overall accuracy of this model was lower for out of domain texts (i.e., second language speech; $F1 = 0.936$) than for in domain texts ($F1 = 0.973$).

Effects of the Amount of Training Data

The amount of data used to train a parsing or tagging model will also affect its accuracy. Models that utilize more training data will (all other variables being equal) be more accurate than those that utilize less training data. Pre-trained models (e.g., BERT, Devlin et al., 2019, pp. 4171–4186; RoBERTa, Liu et al., 2019) help to mitigate this effect because they rely on linguistic parameters derived from very large corpora which are subsequently fine-tuned on training data. This has allowed for more accurate models to be created with less training data. Nonetheless, the general positive relationship between training-set size and downstream tagging/parsing performance remains even when using these pre-trained models (e.g., Pérez-Mayos et al., 2021).

Effects of the Machine Learning Model Used

The machine learning model used will affect a tagger or parser's accuracy. Some well-known taggers and parsers rely on machine learning models that account for a limited amount of contextual information (e.g., 2–3 tokens prior to a target token) and/or classification algorithms (e.g., random forests) that are not as effective as current models. In contrast, transformer-based methods (e.g., BERT Devlin et al., 2019; RoBERTa, Liu et al., 2019), which learn contextualized word-embeddings, account for broader contexts (e.g., an entire sentence), are pre-trained on massive corpora, and are fine-tuned for specific tasks have achieved particularly high performance in automated annotation tasks (e.g., Honnibal et al., 2020; Kyle & Eguchi, 2024; Nguyen et al., 2021).

Other Types of Automatic Linguistic Annotation

POS tagging and syntactic parsing have been common areas of interest in computational and corpus linguistic research. However, automated annotation of features beyond POS tags and syntactic relationships/labels is becoming increasingly common. The advent of pre-trained machine learning models has made reasonably accurate annotation of linguistic features possible with a relatively small amount of training data. For example, Eguchi and Kyle (2023, pp. 429–442) introduced a model that annotates features of authorial stance (Martin & White, 2003) inspired by systemic functional linguistics. Using a pre-trained model, Eguchi and Kyle's model outperformed human annotators using a treebank consisting of only 4,688 sentences. Sung and Kyle (2024) developed an accurate ($F1 = 0.092$) automated tagger for argument structure constructions (ASCs, Goldberg, 1995) using a pre-trained model. Among other uses, this model allows researchers to analyze grammatical development and proficiency in large learner corpora from a usage-based constructionist perspective (e.g., Sung & Kyle, 2025).

Challenges in Tagging and Parsing Low-Resource and Learner Language

In the previous section, it was noted that contemporary taggers/parsers generally achieve high accuracy, especially when encountering data that is similar to the data they were trained on. However, when these models encounter data domains lacking the dense, high-quality annotations essential for supervised learning, their performance degrades substantially. This scenario is known as a “low-resource” setting, which encompasses any language-use domain with insufficient or unrepresentative training data—such as minority dialects, specialized registers, and learner varieties—rather than solely under-resourced national languages (Joshi et al., 2020).

Learner-produced texts, which often exhibit unique vocabulary and atypical grammatical structures that diverge from advanced language-use norms, can be considered a low-resource setting. Consequently, off-the-shelf models frequently mislabel novel words, misparse nonstandard morphological variants, and mishandle unconventional constructions, triggering errors that undermine downstream linguistic analyses (Meurers & Dickinson, 2017). This shortcoming is especially problematic when the research goal is to describe the characteristics of learner language itself. Second language researchers have highlighted the need for language learner specific resources such as annotated datasets (Kyle, 2021; Meurers & Dickinson, 2017).

Recent years have seen growing efforts to alleviate the annotation bottleneck in L2 research through the release of publicly available L2 treebanks, all of which adhere to the UD framework and are hosted in the UD repository. For example, the Treebank of Learner English (TLE; Berzak et al., 2016), comprises approximately 5,000 sentences drawn from intermediate learners' essays in the Cambridge ESOL First Certificate in English (FCE) exam. Extending coverage to spoken learner language, the Spoken L2 English Treebank (SL2E; Kyle et al., 2022b) includes 7,412 utterances extracted from the NICT Japanese Learner English corpus. Beyond English, UD Italian-Valico (Di Nuovo et al., 2022) presents 6,784 tokens elicited from a comic strip by learners of Italian whose first languages include English, French, German, and Spanish. Most recently, UD Korean-KSL (Sung & Shin, 2025) offers 12,984 manually annotated sentences produced by learners of Korean, with each sentence coded for morpheme segmentation, Korean-specific POS tags, and dependency relations. These resources allow for more accurate POS and syntactic annotation of learner texts (e.g., Kyle & Eguchi, 2024), making large corpus analyses of learner language feasible.

Conclusion

Automatic POS tagging and syntactic parsing allow for detailed corpus linguistic analysis to be conducted at scale. Modern tools such as those based on transformer architectures achieve high performance for many annotation types and across many registers. However, tagger and parser accuracy may not be equally high for across all annotations and all registers. It is therefore important for researchers to check annotation accuracy for their particular research purposes.

See Also: Corpus Formats, Markup and Annotation; Learner Corpora; LLMs and Corpora; Pragmatic Annotation.

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